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Case Scenario : MySQL Queries Implementation for Train Ticket Reservation System to Receive Tickets through Social Network

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mysql>
mysql> -- EXPERIMENT 23: Case Scenario - Train Ticket Reservation via Social Network
mysql> DROP DATABASE IF EXISTS train_db;
Query OK, 3 rows affected (0.05 sec)

mysql> CREATE DATABASE train_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE train_db;
Database changed
mysql>
mysql> CREATE TABLE Train (
    ->     train_no INT PRIMARY KEY,
    ->     train_name VARCHAR(50) NOT NULL,
    ->     source VARCHAR(30) NOT NULL,
    ->     destination VARCHAR(30) NOT NULL,
    ->     total_seats INT NOT NULL
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Passenger (
    ->     passenger_id INT PRIMARY KEY,
    ->     name VARCHAR(50) NOT NULL,
    ->     age INT,
    ->     gender VARCHAR(10),
    ->     social_platform VARCHAR(20),
    ->     social_handle VARCHAR(50) UNIQUE
    -> );
Query OK, 0 rows affected (0.04 sec)
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mysql>
mysql> CREATE TABLE Booking (
-> booking_id INT AUTO_INCREMENT PRIMARY KEY,
-> passenger_id INT,
-> train_no INT,
-> booking_date DATE NOT NULL,
-> seat_no INT,
-> status VARCHAR(20) DEFAULT 'CONFIRMED',
-> FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id),
-> FOREIGN KEY (train_no) REFERENCES Train(train_no)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO Train VALUES
-> (1001,'Chennai Express','Chennai','Bengaluru',300),
-> (1002,'Coastal Rail','Chennai','Mumbai',250),
-> (1003,'Southern Star','Madurai','Chennai',200);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Passenger VALUES
-> (1,'Ravi Kumar',28,'Male','WhatsApp','ravi_wa01'),
-> (2,'Priya Sundar',24,'Female','Telegram','priya_tg02'),
-> (3,'Mohammed Ali',35,'Male','WhatsApp','ali_wa03'),
-> (4,'Lakshmi Narayan',30,'Female','Instagram','lakshmi_ig04');
Query OK, 4 rows affected (0.00 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Booking (passenger_id, train_no, booking_date, seat_no, status) VALUES
-> (1,1001,'2026-08-10',22,'CONFIRMED'),
-> (2,1002,'2026-08-11',45,'CONFIRMED'),
-> (3,1001,'2026-08-10',23,'CONFIRMED'),
-> (4,1003,'2026-08-12',10,'WAITLIST');
Query OK, 4 rows affected (0.00 sec)
Records: 4 Duplicates: 0 Warnings: 0

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mysql>
mysql> SELECT b.booking_id, p.name, p.social_platform, t.train_name, b.seat_no, b.status
-> FROM Booking b
-> JOIN Passenger p ON b.passenger_id = p.passenger_id
-> JOIN Train t ON b.train_no = t.train_no;
+-----+-----+-----+-----+-----+-----+
| booking_id | name          | social_platform | train_name    | seat_no | status    |
+-----+-----+-----+-----+-----+-----+
| 1          | Ravi Kumar    | WhatsApp        | Chennai Express | 22      | CONFIRMED |
| 3          | Mohammed Ali  | WhatsApp        | Chennai Express | 23      | CONFIRMED |
| 2          | Priya Sundar  | Telegram        | Coastal Rail    | 45      | CONFIRMED |
| 4          | Lakshmi Narayan | Instagram       | Southern Star   | 10      | WAITLIST  |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT t.train_name, COUNT(*) AS bookings
-> FROM Booking b JOIN Train t ON b.train_no = t.train_no
-> GROUP BY t.train_name;
+-----+-----+
| train_name | bookings |
+-----+-----+
| Chennai Express | 2 |
| Coastal Rail    | 1 |
| Southern Star   | 1 |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT name, social_handle FROM Passenger WHERE social_platform = 'WhatsApp';
+-----+-----+
| name          | social_handle |
+-----+-----+
| Ravi Kumar    | ravi_wa01     |
| Mohammed Ali  | ali_wa03      |
+-----+-----+
2 rows in set (0.00 sec)

mysql>

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mysql>
mysql> UPDATE Booking SET status = 'CANCELLED' WHERE booking_id = 4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
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```
mysql> SELECT * FROM Booking;
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booking_id	passenger_id	train_no	booking_date	seat_no	status
1	1	1001	2026-08-10	22	CONFIRMED
2	2	1002	2026-08-11	45	CONFIRMED
3	3	1001	2026-08-10	23	CONFIRMED
4	4	1003	2026-08-12	10	CANCELLED

```
4 rows in set (0.00 sec)
```